

Highlights

Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study

Julio Azancort

- Recursive degradation is governed by effective training pressure, not adapter rank alone
- A sharp backbone-dependent boundary separates homeostatic and degradative regimes
- Reducing synthetic exposure by approximately 5% restores near-homeostatic stability across three independent random masks
- Low-rank QLoRA reduces one-time adaptation cost versus full fine-tuning by approximately 5 percentage points
- Output drift (verbosity, efficiency loss) serves as a diagnostic signature of above-threshold regimes

Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study

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ARTICLE INFO

Keywords:

recursive synthetic training
model collapse
parameter-efficient fine-tuning
LoRA
effective training pressure
knowledge retention

ABSTRACT

Recursive training on synthetic data causes model collapse, but the transition from stability to degradation is sharper and more controllable than previously understood. We show that recursive factual degradation under QLoRA is governed by effective training pressure, a composite of adapter capacity, update magnitude, and synthetic exposure. Through systematic dose-response mapping across three backbones (Qwen 2.5 1.5B, Gemma 3 1B, Gemma 4 E2B), full fine-tuning comparison, and causal intervention experiments, we find: (1) a sharp regime transition between homeostatic retention and progressive degradation; (2) backbone-dependent thresholds not reducible to a universal effective-rank value; (3) a one-time adaptation cost advantage of approximately 5 percentage points for QLoRA over full fine-tuning at matched perturbation; (4) output drift as a diagnostic signature of above-threshold regimes; and (5) restoration of near-homeostatic stability by reducing synthetic exposure by approximately 5%, confirmed across three independent random masks. These results position recursive degradation as pressure-gated, with practical implications for monitoring and governing recursive training pipelines.

1. Introduction

2. Related Work

3. Methodology

We study the stability of recursive synthetic fine-tuning through the lens of *effective training pressure*: the combined effect of update capacity, perturbation magnitude, and synthetic-data exposure applied at each recursive generation. We do not treat this quantity as a single fitted scalar. Instead, we use it as an organizing framework in which each component is manipulated separately: adapter rank controls update capacity, learning rate controls perturbation magnitude, and filtering or downsampling controls synthetic exposure. This design lets us ask whether transitions between homeostatic and degradative regimes are attributable to capacity, magnitude, exposure, or their interaction.

Our experiments follow a dose-response design across these three axes, tracking factual retention and output-distribution drift over up to ten recursive generations. In the main protocol, each generation is trained only on synthetic outputs produced by the previous generation, with no real-data accumulation, replay buffer, or quality filter. This strict replace-only setting follows the recursive data-replacement paradigm studied in the model-collapse literature (Shumailov, Shumaylov, Zhao, Papernot, Anderson and Gal, 2024; Dohmatob, Feng, Subramonian and Kempe, 2025; Gerstgrasser, Schaeffer, Dey, Rafailov, Pai, Sleight, Hughes, Korbak, Agrawal, Gromov, Roberts, Yang, Donoho and Koyejo, 2024), while allowing us to isolate how the update mechanism changes the resulting dynamics.

The design is motivated by both practical and theoretical considerations. In practice, parameter-efficient fine-tuning

via LoRA (Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang and Chen, 2022) and QLoRA (Detrmers, Pagnoni, Holtzman and Zettlemoyer, 2023) dominates production adaptation of large language models, yet behavior under recursive synthetic exposure remains poorly characterized beyond the observation that LoRA can “learn less and forget less” (Biderman, Portes, Ortiz, Paul, Greengard, Jernite, Kolber, Teng, Schenck et al., 2024). That work does not map when this regularization remains protective and when it fails under repeated synthetic training. Theoretically, fine-tuning updates often lie in low-dimensional subspaces (Aghajanyan, Gupta and Zettlemoyer, 2021), suggesting that adapter rank directly modulates the effective dimensionality of the perturbation applied each generation. We operationalize this via the effective rank of the composed update matrices BA , following the entropy-based definition of Roy and Vetterli (2007).

The remainder of this section is organized as follows: the recursive protocol (§3.1), factual retention benchmark (§3.2), model backbones (§3.3), experimental axes (§3.4), effective-rank and perturbation metrics (§3.5), output-distribution diagnostics (§3.6), synthetic-exposure interventions (§3.7), and statistical conventions (§3.8).

3.1. Recursive replace-only protocol

Our main recursion protocol is *data-recursive* rather than *weight-recursive*. At each generation, the training data are replaced by synthetic outputs produced by the previous generation, but model updates are not accumulated across generations. In the QLoRA experiments, the base model remains frozen and a fresh adapter is initialized at each generation. In the full fine-tuning (FFT) experiments, the same replace-only data recursion is used, but the full model weights are updated instead of a low-rank adapter. This design separates degradation in the synthetic data stream from artifacts caused by adapter stacking, LoRA merging, or cumulative checkpoint continuation.

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Let $\mathcal{Q}_{\text{train}}$ denote the set of training questions used to generate synthetic data. At generation 0, the base model answers $\mathcal{Q}_{\text{train}}$ to produce the initial synthetic dataset S_0 . For each subsequent generation $t \in \{1, \dots, T\}$, the protocol proceeds as follows:

1. A new trainable update mechanism is initialized: a fresh QLoRA adapter for PEFT experiments, or a fresh full fine-tuning run for FFT experiments.
2. The model is fine-tuned for 2 epochs on S_{t-1} , using standard causal language modeling loss over the formatted prompt-response sequences.
3. The resulting model is evaluated on a held-out factual retention set described in §3.2.
4. The fine-tuned model generates a new synthetic dataset S_t by answering the same training questions $\mathcal{Q}_{\text{train}}$ using stochastic decoding.
5. The trained update and optimizer state are not carried forward. Only the newly generated dataset S_t is used to train the next generation.

Unless otherwise stated, the synthetic training stream consists of 2,000 questions sampled from the TriviaQA rc.nocontext split (Joshi, Choi, Weld and Zettlemoyer, 2017), shuffled with fixed seed 15. A disjoint set of 200 questions is held out for factual retention evaluation. The held-out set is never used for synthetic-data generation or training. Synthetic responses are generated in batches of 8 using the model-specific chat template, with stochastic decoding (temperature 0.7, top- $p = 0.9$, maximum 30 new tokens). The resulting prompt-response sequences are stored and used as the training corpus for the next generation.

For QLoRA runs, adapters are trained with learning rate 10^{-5} , effective batch size 16 (micro-batch size 2 with 8 gradient-accumulation steps), bf16 precision, LoRA scaling $\alpha = 2r$, dropout 0.05, and maximum sequence length 256 tokens. Unless otherwise noted, no warmup, weight decay, or learning-rate scheduling is applied. FFT runs use the same recursive data protocol and evaluation procedure, but update the full model weights; FFT-specific optimizer and learning-rate settings are described in §3.4.

The protocol is intentionally replace-only: previous real or synthetic datasets are not accumulated, and no replay buffer is used. This contrasts with accumulation-based mitigation strategies (Gerstgrasser et al., 2024) and with loss-level interventions such as truncated cross-entropy losses (Zibakhah, Shohgahi, Aghazadeh, Mirhoseini and Koushanfar, 2024). We use this strict setting to expose how recursive stability changes as a function of effective training pressure. Controlled interventions on synthetic exposure, including filtering and downsampling, are introduced separately in §3.7.

3.2. K0 factual retention metric

Our primary evaluation metric is *K0 retention*: the fraction of facts that the base model answers correctly at generation 0 and continues to answer correctly at generation t . This metric isolates factual *loss* from factual *absence*: models are

evaluated only on facts they demonstrably possessed at the start of the recursive process.

At generation 0, the base model without any adapter is evaluated on a held-out set of 200 TriviaQA questions. Each question includes multiple valid answer aliases. A response is scored as correct if, after normalization, the predicted string contains any ground-truth alias or any ground-truth alias contains the predicted string. This bidirectional substring criterion is intended to handle short-answer variation and aliasing in TriviaQA-style factoid responses. Normalization consists of lowercasing, removing articles (*the*, *a*, *an*), stripping punctuation, converting written number words to digits, and collapsing whitespace.

The subset of held-out questions answered correctly at generation 0 defines K_0 . For example, for Qwen 2.5 1.5B, $|K_0| = 78$ out of 200 held-out questions. Other backbones have their own fixed K_0 sets, determined once at generation 0 and held constant across all experimental conditions for that backbone.

At each subsequent generation t , only the corresponding K_0 items are used to compute retention:

$$\text{Retention}(t) = \frac{|\{q \in K_0 : q \text{ is answered correctly at generation } t\}|}{|K_0|}. \quad (1)$$

Thus, a retention score of 100% means that the model preserved all factual items it answered correctly at generation 0. The metric does not measure global factual accuracy, nor does it reward acquisition of facts outside K_0 ; it is designed specifically to measure preservation of initially demonstrated knowledge under recursive training.

We additionally track item-level transitions within K_0 . A correct-to-wrong transition ($C \rightarrow W$) indicates that an initially retained fact was lost at a later generation. A wrong-to-correct transition ($W \rightarrow C$) indicates that a previously lost K_0 item became correct again in a later generation. These transition counts help distinguish monotonic forgetting, temporary instability, and bounded oscillation even when aggregate retention is similar.

Because the same normalization and alias-matching procedure is applied at every generation and to every condition within a backbone, our analysis emphasizes relative retention dynamics rather than absolute TriviaQA benchmark performance. We report K_0 retention as both a fraction and percentage (e.g., $73/78 = 93.6\%$). For multi-seed experiments, we report per-seed values and summaries across seeds. Unless otherwise stated, K_0 is defined separately for each backbone and then reused unchanged across rank, learning-rate, module targeting, and intervention conditions.

3.3. Models and hardware

We conduct experiments on three small instruction-tuned transformer backbones in the 1–2B parameter range, spanning two model families. Each backbone serves a different evidential role: Qwen2.5-1.5B is the primary model for detailed dose-response analysis, Gemma 3 1B is used for cross-architecture threshold validation, and Gemma 4 E2B

is included as an additional single-seed robustness check. Architectural metadata are taken from the released model configurations unless otherwise stated.

- **Qwen2.5-1.5B-Instruct** (Qwen Team, 2024) is the primary backbone used for the QLoRA rank sweep, full fine-tuning comparison, distributional-output analysis, and synthetic-exposure interventions. The model has 1.54B parameters, 28 transformer layers, hidden size 1536, and grouped-query attention with 12 attention heads and 2 key-value heads.
- **Gemma 3 1B IT** (Gemma Team, 2025) is the secondary backbone used for testing whether the regime transition generalizes outside the Qwen family. The released model configuration specifies 26 layers, hidden size 1152, and grouped-query attention with 4 attention heads and 1 key-value head. We evaluate the boundary between homeostatic and degradative regimes with three seeds at the relevant ranks ($r = 4$ and $r = 16$).
- **Gemma 4 E2B IT**¹ is used as an additional single-seed robustness check. We use it to test whether low-effective-rank recursive fine-tuning remains stable in a third backbone, but we do not use it to estimate a degradation threshold.

Unless otherwise stated, QLoRA experiments load the base model in 4-bit NormalFloat (NF4) quantization with double quantization and bfloat16 compute dtype, following the QLoRA protocol (Dettmers et al., 2023). Full fine-tuning experiments train the model weights in bfloat16 without 4-bit weight quantization; memory usage is reduced through gradient checkpointing and an 8-bit paged optimizer rather than model quantization.

Experiments were run on a single NVIDIA RTX 4000 Ada GPU with 20 GB of VRAM. In our implementation, a single QLoRA generation on Qwen2.5-1.5B completes in a few minutes at low ranks and approximately 12 minutes at $r = 256$; full fine-tuning generations require approximately 25 minutes with gradient checkpointing. A complete 10-generation QLoRA recursive run at $r = 16$ completes in under one hour. These timings are reported to document that the full protocol is reproducible on a single workstation-class GPU.

All code, configuration files, and experimental outputs are publicly available at <https://github.com/Jazancort/llm-knowledge-collapse>.

3.4. Experimental axes

Our dose-response design manipulates effective training pressure along five experimental axes. Each axis targets a different component of pressure while keeping the recursive protocol, dataset, and evaluation procedure fixed. The axes are not intended to define a single scalar pressure value;

rather, they isolate how different sources of pressure (update capacity, perturbation magnitude, module topology, and synthetic exposure) affect recursive stability. Table 1 summarizes the experimental axes, the manipulated variable, and the tested range.

Table 1

Experimental axes and their corresponding pressure components.

Axis	Manipulated variable	Range
1. Update capacity	QLoRA rank	$r \in \{4, 16, 32, 64, 128, 256\}$
2. Perturbation magnitude	FFT learning rate	$\{10^{-6}, 5 \times 10^{-6}, 10^{-5}, 2 \times 10^{-5}\}$
3. Method comparison	QLoRA vs FFT	Approx. perturbation
4. Module topology	Target modules	Attention-only vs full
5. Synthetic exposure	Training examples	100% vs ~95% filter

Axis 1: Adapter rank (update capacity). We vary QLoRA rank across six values ($r = 4, 16, 32, 64, 128, 256$) on the Qwen backbone, producing adapters ranging from 545K to 34.9M trainable parameters. All other QLoRA hyperparameters are held fixed. This axis tests whether increasing the dimensionality of the per-generation update shifts the system from homeostatic to degradative behavior. The headline ranks ($r = 16$ and $r = 256$) are evaluated with three seeds; intermediate ranks are evaluated with a single seed and used to trace the dose-response curve. On Gemma 3, we test $r \in \{2, 4, 16, 256\}$, with three seeds at the boundary ranks ($r = 4$ and $r = 16$).

Axis 2: Learning rate (perturbation magnitude). For full fine-tuning, we sweep the learning rate across four values ($10^{-6}, 5 \times 10^{-6}, 10^{-5}, 2 \times 10^{-5}$) on Qwen while keeping the recursive data protocol fixed. This produces per-generation weight drifts ranging from 0.39 to 3.48, measured as the mean absolute parameter change. This axis tests whether perturbation magnitude can drive the homeostatic-to-degradative transition independently of adapter rank.

Axis 3: Method comparison (QLoRA vs FFT). We compare QLoRA $r = 16$ against FFT at LR = 10^{-6} . This FFT learning rate was selected because it produces the closest available perturbation proxy to the QLoRA adapter update: the effective LoRA update norm $\|BA\|_F \approx 0.42$ and the FFT weight drift is approximately 0.39. Because these quantities are not identical geometric objects, we describe this as an *approximately perturbation-matched* comparison. Both methods are run for 10 generations with three seeds, enabling comparison of one-time adaptation cost and long-term stability under low-perturbation recursive training.

Axis 4: Module topology (supportive ablation). On Qwen, we compare updating attention modules only (q_proj, v_proj) with updating all linear modules (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). This ablation tests whether recursive stability is primarily explained by where the adapter is placed or by the effective capacity of the update. Because these module-targeting runs are evaluated

¹Available at <https://huggingface.co/google/gemma-4-e2b-it>. No formal technical report has been published at time of writing.

with a single seed, they serve as supportive evidence rather than headline claims.

Axis 5: Synthetic exposure (intervention). We manipulate the synthetic training stream in the high-capacity Qwen $r = 256$ boundary regime to test whether reducing effective synthetic exposure can shift the system back toward homeostasis. Interventions include length filtering (removing responses longer than 5 words), random downsampling with three independent masks removing approximately 5% of examples, and short-answer prompt constraining. These interventions test exposure sensitivity rather than serving as a general data-filtering recipe. The intervention design is detailed in §3.7.

3.5. Effective training pressure

We use *effective training pressure* as a conceptual framework for interpreting recursive fine-tuning stability, rather than as a single fitted equation. The term refers to the effective perturbation imposed on the model and the synthetic training stream at each recursive generation. In our experiments, this pressure arises from three jointly contributing factors:

1. **Update capacity:** the dimensionality of the trainable update subspace, controlled by QLoRA rank and measured through effective rank.
2. **Perturbation magnitude:** the size of the parameter update induced by one generation of training, controlled by learning rate in FFT and approximated by the norm of the composed LoRA update in QLoRA.
3. **Synthetic exposure:** the amount and composition of model-generated training data fed back into the next generation, manipulated through filtering or downsampling of the synthetic stream.

We do not assume a known functional form combining these components. Instead, our experiments manipulate each component separately and show that changes along any of the three axes can move the system between bounded and degradative regimes. The framework is therefore empirically motivated: it summarizes a set of observed control knobs rather than proposing a closed-form theory of recursive collapse.

Effective rank as a within-backbone capacity measure. To quantify update capacity beyond nominal rank, we compute the effective rank of the composed LoRA update matrices BA after training at each generation. Following Roy and Vetterli (2007), the effective rank is defined as

$$\text{erank}(BA) = \exp\left(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\right), \quad (2)$$

where $\hat{\sigma}_i = \sigma_i / \sum_j \sigma_j$ are the normalized singular values of BA . This measure captures how many singular directions carry substantial weight in the adapter update. A rank- r adapter with uniformly distributed singular values reaches

$\text{erank} = r$, whereas an update dominated by a single singular direction has $\text{erank} \approx 1$ regardless of nominal rank.

Unless otherwise stated, we report the mean effective rank across adapted matrices in a model. For comparisons across module-targeting schemes, we also consider aggregate effective rank, defined as the sum of effective ranks across adapted matrices. In our within-backbone rank sweeps, effective rank generally increases with nominal rank and organizes the observed transition between homeostatic and degradative regimes better than nominal rank alone. However, raw effective-rank thresholds are not portable across architectures: Qwen remains bounded at substantially higher effective ranks than Gemma 3. We therefore treat effective rank as a within-backbone capacity measure and discuss cross-architecture normalization as a limitation in §6.

Perturbation magnitude for full fine-tuning. For full fine-tuning, we measure perturbation magnitude as the mean absolute parameter change relative to the original model weights:

$$\text{drift}(t) = \frac{1}{|\theta|} \sum_i |\theta_i^{(t)} - \theta_i^{(0)}|. \quad (3)$$

Here $\theta^{(0)}$ denotes the original model parameters and $\theta^{(t)}$ the parameters after one generation of fine-tuning. This scalar summarizes the average displacement of the model in weight space under FFT. For QLoRA, the closest available perturbation proxy is the Frobenius norm of the composed adapter update, $\|BA\|_F$. These two quantities are not geometrically identical: one measures displacement of the full model weights, while the other measures the magnitude of a low-rank update. We therefore use them only as approximate perturbation proxies when selecting the FFT/QLoRA comparison in Axis 3.

3.6. Output-distribution diagnostics

Beyond factual retention, we track output-distribution diagnostics designed to measure how the synthetic training stream changes across recursive generations. These metrics serve two purposes. First, they detect distributional degradation that may not be captured by K0 retention alone. Second, they provide monitorable signals that could be used to audit recursive fine-tuning pipelines. We treat these diagnostics as descriptive indicators of output drift, not as standalone causal explanations for factual loss.

Content efficiency. We define content efficiency as factual retention normalized by mean response length:

$$\text{ContentEff}(t) = \frac{\text{Retention}(t)}{\text{MeanLength}(t)}. \quad (4)$$

Here, $\text{MeanLength}(t)$ is measured in words over the generated synthetic responses at generation t . This composite metric captures whether factual retention is being maintained with increasingly verbose outputs. Declining content efficiency indicates that the model is producing more tokens

per retained fact, whereas stable content efficiency indicates that retention and response length remain proportionally stable.

Baseline response persistence. For each K_0 item, we compare the model's normalized response at generation t with its normalized response at generation 0. The strict persistence score is

$$\text{Persistence}(t) = \frac{|\{q \in K_0 : \text{answer}_t(q) = \text{answer}_0(q)\}|}{|K_0|} \quad (5)$$

High persistence indicates that responses remain anchored to the model's generation-0 outputs. Declining persistence indicates response drift relative to the initial model behavior, even when aggregate factual retention remains bounded. Because exact matching can be sensitive to minor wording changes, we also compute fuzzy item-level stability using token-overlap similarity as a robustness check.

Lexical diversity and verbosity. We measure lexical diversity of the synthetic training stream using several corpus-level metrics. Distinct- n (Li, Galley, Brockett, Gao and Dolan, 2016) computes the ratio of unique n -grams to total n -grams in the generated corpus; we report Distinct-1 and Distinct-2. Because Distinct- n is sensitive to response length, we also compute the Measure of Textual Lexical Diversity (MTLD; McCarthy and Jarvis, 2010), a length-robust estimate of lexical diversity. In addition, we track mean response length and stopword ratio, which help distinguish verbosity increase from changes in content-word diversity.

Synthetic Drift Index (SDI-3). To summarize output-distribution change, we define a three-term synthetic drift index:

$$\text{SDI-3} = \log\left(\frac{\text{MeanLength}_T}{\text{MeanLength}_0}\right) + \log\left(\frac{\text{Distinct}1_0}{\text{Distinct}1_T}\right) + \Delta\text{Instability}, \quad (6)$$

where $\Delta\text{Instability} = \text{Instability}(T) - \text{Instability}(1)$ and $\text{Instability}(t)$ denotes the fraction of training items whose normalized response differs between generation $t-1$ and generation t . The three terms correspond to verbosity increase, unigram diversity decrease, and loss of item-level response stability. SDI-3 is not proposed as a theoretical measure of collapse; it is an empirical summary statistic used to compare output drift across experimental conditions.

These diagnostics complement K_0 retention by measuring distributional properties of the synthetic stream. K_0 retention measures preservation of initially known facts, whereas the output diagnostics measure whether the generated training data remain short, lexically efficient, and anchored to the model's initial responses. Throughout the paper, we interpret these metrics as signatures of distributional drift. Causal claims are reserved for explicit interventions on the synthetic training stream in §3.7.

3.7. Synthetic-exposure interventions

The interventions in Axis 5 test whether the degradative regime observed in the high-capacity Qwen $r = 256$ condition can be shifted back toward homeostasis by modifying the synthetic training stream. All interventions use the same backbone, rank, dataset, and recursive protocol as the Qwen $r = 256$ baseline. The initial intervention comparison is conducted with seed 15 for 5 generations; the strongest intervention is then replicated across additional seeds or random masks, as described below.

We define five conditions:

- **C1 (Normal):** the unmodified baseline. Synthetic responses are generated and used for training without filtering, downsampling, or post-processing. This is the same replace-only protocol used in the rank sweep.
- **C2 (Short-answer constrained):** the generation prompt is modified to request a brief factual answer, and the maximum generation length is reduced to 10 tokens. This constrains output verbosity at generation time while retaining all training examples.
- **C3 (Length-filtered):** synthetic responses are generated with the normal prompt, then examples whose responses exceed 5 words are removed from the next-generation training set. The remaining examples are used without duplication.
- **C4 (Canonical extraction):** synthetic responses are generated normally, then a heuristic extracts a short factoid span from each response and replaces the original response with this canonicalized answer.
- **C5 (C3-matched random downsampling):** synthetic responses are generated normally, then a random subset of examples is selected so that the resulting training stream approximately matches the token budget of C3. The subset is chosen by a random permutation seeded independently of the training seed.

The key comparison is between C3 and C5. C3 reduces exposure by removing long responses, whereas C5 reduces exposure by removing a random subset of examples until the training stream matches C3's approximate token budget. Thus, if both conditions produce similar stabilization, the effect cannot be attributed solely to removing long responses; it instead suggests sensitivity to marginal synthetic exposure in this boundary regime.

To test robustness, C3 is evaluated with three training seeds (15, 137, 256). C5 is evaluated with three independent random downsampling masks (15, 42, 99). We compute Jaccard overlap between the examples removed by C3 and by each C5 mask to verify that the interventions remove largely distinct subsets of the training stream.

C2 and C4 provide additional controls on output format. C2 changes the generation instruction and length limit while retaining all examples; C4 canonicalizes responses after generation. These conditions help distinguish exposure

reduction from response-format control. We report token counts, example counts, and retention for all intervention conditions to assess whether improvements can be explained by reduced training volume alone.

3.8. Seeds, statistics, and reporting

Recursive fine-tuning experiments are computationally expensive because each condition requires multiple generations of training and data regeneration. We therefore use a small but explicit seed policy, distinguishing between multi-seed headline comparisons and single-seed supporting ablations.

Seed policy. Headline comparisons are evaluated with three seeds (15, 137, 256). These include the Qwen rank-sweep endpoints ($r = 16$ and $r = 256$), the Gemma 3 boundary ranks ($r = 4$ and $r = 16$), the QLoRA-FFT comparison, and the C3/C5 synthetic exposure interventions. Intermediate dose-response points and supporting ablations, including module-topology comparisons, Gemma 4 robustness checks, and single-rank FFT learning-rate sweeps, are evaluated with a single seed and labeled as such. We do not pool single-seed and multi-seed results when making headline claims.

Reporting format. For three-seed conditions, we report per-seed values together with the mean and observed range. Because $N = 3$ provides imprecise estimates of sampling variability, we use these summaries descriptively rather than as estimates of population-level uncertainty. When the observed ranges of two conditions do not overlap across all three seeds, we treat this as consistent empirical separation under the present experimental protocol, not as a formal hypothesis test.

Statistical approach. We avoid formal hypothesis testing. With only three seeds per condition, classical tests such as the Wilcoxon signed-rank test have very limited statistical power. We therefore emphasize effect sizes, monotonic dose-response behavior, and per-seed consistency. Our primary evidential pattern is ordinal and mechanistic: retention changes systematically as update capacity, perturbation magnitude, or synthetic exposure is varied. In particular, the QLoRA rank sweep and FFT learning-rate sweep are interpreted as dose-response curves across six and four levels respectively, where the ordinal consistency of the entire sweep provides stronger evidence than any single pairwise comparison.

Effect sizes. We report absolute differences in factual retention as percentage-point gaps. Effect sizes are emphasized because they remain directly interpretable regardless of sample size. For QLoRA-FFT comparisons, gaps are reported per seed and as a mean. For synthetic-exposure interventions, we report the improvement over the unmodified C1 baseline at the same generation. Where relevant, we also report item-level transition counts and the number of

K_0 facts lost or regained. We do not claim precise threshold locations or a universal effective-rank value at which degradation begins; instead, we identify the interval between adjacent tested settings where the observed regime transition occurs. Single-seed observations are used as exploratory or supportive evidence and are not used as standalone support for headline conclusions.

Together, the subsections above define a methodology structured around five complementary experimental axes, a factual retention metric that isolates knowledge loss from knowledge absence, a suite of distributional diagnostics for detecting output-distribution drift beyond retention, and a set of controlled synthetic-exposure interventions for testing boundary sensitivity. The results section is organized to mirror this design: we first establish the capacity-gated regime transition through the Qwen rank sweep and Gemma 3 cross-backbone validation (Axes 1–2), then compare QLoRA and full fine-tuning at approximately matched perturbation (Axis 3), characterize distributional signatures accompanying degradation, and finally show that marginal exposure reduction restores near-homeostatic retention at the boundary (Axis 5). Module topology ablations (Axis 4) are presented as supporting evidence alongside the capacity results.

4. Results

The results are organized by experimental axis to examine how the principal components of effective training pressure relate to recursive stability and degradation. We begin with the central empirical result: a clear, monotonic regime transition as adapter rank increases on the Qwen backbone (§4.1). We then evaluate whether the same behavior generalizes across different model families (§4.2), investigate whether increasing perturbation magnitude through full fine-tuning reproduces the same qualitative pattern (§4.3), characterize the distributional signatures associated with above-threshold regimes (§4.4), and present evidence that a marginal reduction in synthetic exposure restores near-homeostatic behavior at the regime boundary (§4.5). Throughout this section, headline findings are supported by multi-seed experiments, whereas single-seed results are presented as supporting evidence. Consistent with the methodology, comparisons emphasize effect sizes and reproducible experimental patterns rather than formal inferential statistics.

4.1. Update capacity: QLoRA rank dose-response

Increasing adapter rank produces a clear, monotonic decline in factual retention across the full rank sweep on Qwen 2.5 1.5B. Table 2 summarizes K_0 retention, effective rank, trainable parameters, and regime classification for each tested configuration. Across the evaluated range, Gen10 retention decreases from 94.9% at $r = 16$ to 78.0% at $r = 256$, a difference of 16.9 percentage points.

The longitudinal trajectories (Fig. 1) reveal qualitatively different temporal dynamics across regimes. At $r = 16$, retention stabilizes after an initial one-generation adaptation

Table 2

QLoRA rank sweep on Qwen 2.5 1.5B-Instruct. Retention is reported at the last evaluated generation. Headline ranks ($r = 16$ and $r = 256$) are evaluated with three seeds; all others with a single seed. Configurations evaluated only to Gen5 correspond to intermediate dose-response points.

Rank	Params	Gen	Retention	Eff. Rank	Regime
$r = 4$	545K	5	96.2%	3.34	Homeostatic
$r = 16$	2.2M	10	94.9% (N=3)	11.08	Homeostatic
$r = 32$	4.4M	5	94.9%	17.85	Homeostatic
$r = 64$	8.7M	10	91.1%	29.52	Homeostatic
$r = 128$	17.4M	10	88.6%	50.16	Bounded
$r = 256$	34.9M	10	78.0% (N=3)	87.57	Degradative

cost and remains flat through Gen10 (mean 94.9%, range 94.9–97.4% across three seeds). At $r = 128$, retention settles at approximately 88–89% by Gen3 and remains bounded through Gen10, neither recovering nor declining further. This bounded behavior distinguishes it from both the stable homeostatic regime and the progressively degradative regime. At $r = 256$, retention declines progressively at approximately 2 percentage points per generation on average, reaching 78.0% (mean across three seeds) by Gen10 with no sign of stabilization within the observed horizon.

Several features of the dose-response curve (Fig. 2) merit attention. First, the relationship between rank and retention is monotonically decreasing across all six values, spanning more than an order of magnitude in trainable parameters (545K to 34.9M). No rank reversal or non-monotonicity is observed. Second, the decline accelerates at higher ranks: the gap between $r = 64$ (91.1%) and $r = 256$ (78.0%) is substantially larger than any gap between adjacent lower-rank configurations. Third, effective rank increases sub-linearly with nominal rank (3.34 at $r = 4$ to 87.57 at $r = 256$), indicating that higher-rank adapters do not fully utilize their nominal capacity while still producing substantially larger effective updates.

The multi-seed experiments at the rank-sweep endpoints confirm that the regime separation is reproducible. At $r = 16$, per-seed Gen10 retention values are 94.9%, 97.4%, and 97.4%. At $r = 256$, per-seed Gen10 values are 75.9%, 77.2%, and 81.0%. The ranges do not overlap: the lowest $r = 16$ value (94.9%) exceeds the highest $r = 256$ value (81.0%) by 13.9 percentage points, demonstrating clear empirical separation under the present experimental protocol.

As a supportive ablation, we compared attention-only and full-linear module targeting on Qwen at a single seed. Targeting all linear modules (q,k,v,o,gate,up,down) at $r = 4$ produces 4.6M trainable parameters and 94.9% Gen5 retention, matching attention-only targeting at $r = 16$ (2.2M parameters, same retention). The full-linear $r = 4$ adapter has a mean effective rank of 3.43, compared to 11.08 for attention-only $r = 16$. This preliminary observation is consistent with effective update capacity, rather than module location or parameter count alone, being the primary predictor of regime membership. Because it is based on a single seed, we treat it as supportive evidence.

Together, these results establish that increasing update capacity through adapter rank is sufficient to produce a monotonic transition from homeostatic to degradative recursive fine-tuning on the Qwen backbone. The following sections examine whether this transition generalizes across architectures and update mechanisms.

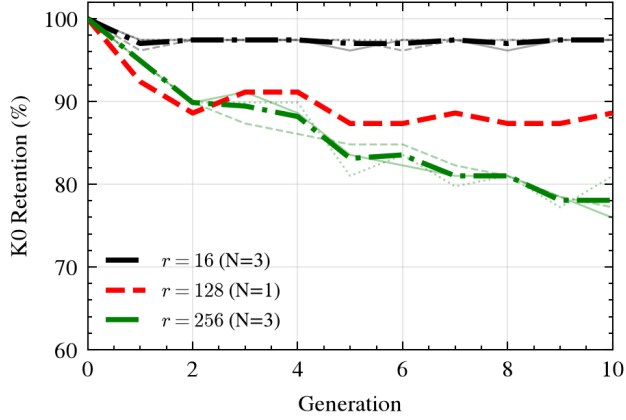


Figure 1: K0 retention trajectories over 10 recursive generations for three representative ranks on Qwen 2.5 1.5B. Thin lines show individual seeds; bold lines show the mean. The $r = 16$ and $r = 256$ conditions are each evaluated with three seeds (15, 137, 256); $r = 128$ is evaluated with a single seed.

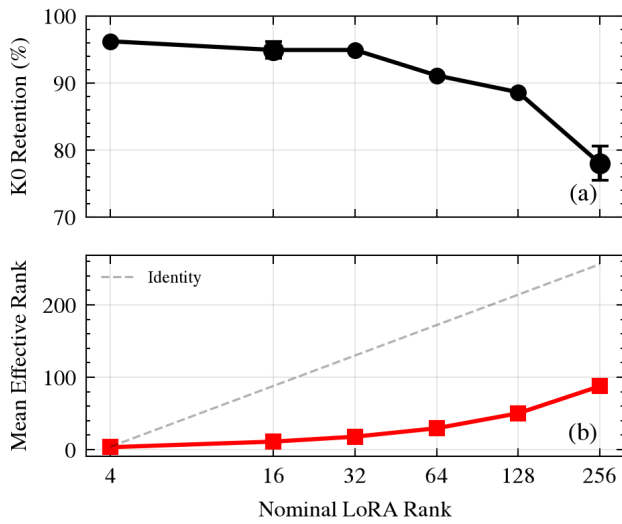


Figure 2: Dose-response relationship between nominal adapter rank and recursive stability on Qwen 2.5 1.5B. Left panel: Gen10 K0 retention (%) versus nominal rank. Right panel: mean effective rank versus nominal rank. Retention declines monotonically with increasing effective update capacity; the sharpest drop occurs between $r = 128$ (bounded, 88.6%) and $r = 256$ (degradative, 78.0%).

4.2. Cross-backbone thresholds

To test whether the regime transition observed on Qwen is architecture-specific, we evaluated corresponding rank configurations on two additional backbones: Gemma 3 1B IT, with three seeds at the boundary ranks, and Gemma 4 E2B IT, with a single seed as a robustness check.

On Gemma 3, the same qualitative structure emerges, but at a substantially lower effective rank. Table 3 summarizes the results. At $r = 4$ (mean effective rank 3.07), Gemma 3 remains homeostatic across three seeds (mean Gen5 retention 92.9%, range 91.5–93.6%). At $r = 16$ (mean effective rank 9.17), Gemma 3 enters the degradative regime across all three seeds (mean Gen5 retention 68.8%, range 68.1–70.2%). The ranges do not overlap, providing consistent empirical separation between the two regimes on this backbone.

Table 3

QLoRA rank sweep on Gemma 3 1B IT. Boundary ranks ($r = 4$ and $r = 16$) are evaluated with three seeds; $r = 2$ and $r = 256$ are evaluated with a single seed. Retention is reported at Gen5 for comparability across ranks; the $r = 4$ condition was additionally extended to Gen10 (seed 15: 93.6%).

Rank	Gen	Retention	Eff. Rank	Regime
$r = 2$	5	97.9%	1.75	Homeostatic
$r = 4$	5	92.9% (N=3)	3.07	Homeostatic
$r = 16$	5	68.8% (N=3)	9.17	Degradative
$r = 256$	5	70.2%	71.47	Degradative

Two additional observations are noteworthy. First, the Gemma 3 threshold lies between effective ranks 3 and 9, substantially below the Qwen threshold (between effective ranks 50 and 88). These results indicate that raw effective-rank values are not directly portable across architectures. Second, Gemma 3 exhibits a post-threshold plateau: $r = 16$ and $r = 256$ reach comparable retention floors (68.8% vs 70.2%), indicating that additional capacity beyond the threshold does not worsen degradation further within the observed horizon. This plateau behavior is observed on Gemma 3; whether the same saturation occurs on other backbones or at different generation depths remains an open question.

On Gemma 4 E2B IT, both tested ranks ($r = 4$, effective rank 2.13; $r = 16$, effective rank 5.64) remain homeostatic at Gen5 (96.1% and 97.4%, respectively). Both configurations fall below the effective-rank threshold observed on Gemma 3, and both remain homeostatic, consistent with the pattern established on Qwen and Gemma 3. We treat this as a robustness confirmation that low-effective-rank adapters can remain stable across multiple backbone architectures, not as an estimate of the Gemma 4 degradation threshold.

We attempted to normalize effective-rank thresholds across architectures by dividing by hidden dimension, number of layers, number of attention heads, and baseline diversity. None of these normalizations reduced the discrepancy to a common threshold. The threshold is therefore likely influenced by architectural properties not captured by simple model-dimension ratios. We report this as a negative result and discuss it further in §6.

Figure 3 visualizes the backbone-specific placement of these thresholds in effective-rank space.

The homeostatic-to-degradative regime transition appears general across the evaluated backbones: it is observed on both Qwen and Gemma 3, with the same qualitative structure (bounded below threshold, degradative above). However, its location in effective-rank space is backbone-dependent. Having established that the regime structure generalizes across architectures, we next examine whether the same transition occurs when perturbation magnitude is varied independently of adapter capacity.

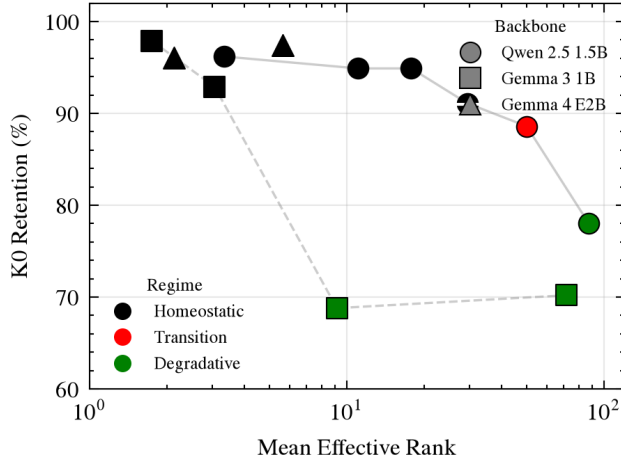


Figure 3: K0 retention versus mean effective rank across three backbones. Marker shapes distinguish backbones (circle: Qwen, square: Gemma 3, triangle: Gemma 4); colors indicate regime membership (blue: homeostatic, orange: transition zone, red: degradative). The regime transition occurs at substantially different effective-rank values across architectures.

4.3. Perturbation magnitude: FFT learning-rate sweep

The previous sections established that increasing update capacity through adapter rank shifts the system from homeostatic to degradative regimes. This section asks whether increasing perturbation magnitude through full fine-tuning, without varying adapter rank, reproduces the same qualitative pattern.

FFT dose-response. We swept learning rate across four values on Qwen 2.5 1.5B under the same recursive replacement-only protocol. Table 4 shows that retention at Gen3 decreases monotonically as learning rate and weight drift increase. Drift values span nearly an order of magnitude (0.39 to 3.48), and retention decreases from 92.3% to 84.6% across this range. These results indicate that perturbation magnitude alone can drive the system toward degradation without any adapter or rank constraint.

Table 4

FFT learning-rate sweep on Qwen 2.5 1.5B (seed 15, Gen3). Perturbation proxy is the mean absolute weight drift relative to the original model for FFT, and $\|BA\|_F$ for QLoRA. These are operationally comparable but not geometrically identical measures. QLoRA $r = 16$ is included as a reference point.

Method	LR	Pert. proxy	Gen3 Retention
QLoRA $r = 16$	10^{-5}	0.42 ($\ BA\ _F$)	96.2%
FFT	10^{-6}	0.39	92.3%
FFT	5×10^{-6}	1.56	91.0%
FFT	10^{-5}	1.95	89.7%
FFT	2×10^{-5}	3.48	84.6%

QLoRA versus FFT at approximately matched perturbation. To compare the two update mechanisms at similar perturbation, we selected FFT at LR = 10^{-6} (drift 0.39) as the closest available match to QLoRA $r = 16$ ($\|BA\|_F \approx 0.42$). Both were run for 10 generations with three seeds. As illustrated in Figure 4, the separation between the trajectories is established after the first recursive generation and remains nearly constant thereafter.

Both methods are stable from Gen1 through Gen10, with no progressive degradation within the observed horizon. QLoRA retains a consistent advantage of approximately 5.6 percentage points (mean Gen10: QLoRA 97.4% vs FFT 91.9%). This gap emerges entirely at Gen1 and does not widen over subsequent generations. We therefore interpret it as a difference in one-time adaptation cost rather than a differential degradation rate: FFT incurs a larger initial factual cost when adapting to the synthetic stream, while both methods exhibit similarly stable trajectories once adapted.

Per-seed Gen10 values for QLoRA are 97.4%, 97.4%, and 97.4%; for FFT they are 92.3%, 91.0%, and 92.3%. The ranges do not overlap across any seed pairing.

Fragile-fact floor analysis. Item-level analysis reveals an asymmetry between the two methods. Under FFT, the

same six K_0 items are lost across all three seeds (cross-seed Jaccard similarity 1.0), indicating a deterministic fragile-fact floor: these facts are consistently displaced by full-parameter updates regardless of seed. Under QLoRA, only two to three facts are lost per seed, and the overlap between facts lost by QLoRA and facts lost by FFT is zero across all seed pairs (Jaccard 0.0). In this low-perturbation regime, low-rank adaptation loses fewer facts and loses different facts than full fine-tuning at comparable perturbation magnitude.

We refer to this pattern as a stable fragile-fact floor under FFT: certain facts are reliably displaced by full-parameter updates, whereas low-rank updates preserve those same facts while occasionally losing others. Whether this reflects different optimization trajectories or structural differences in the update subspace remains an open question for future work.

Both update capacity (Axis 1) and perturbation magnitude (Axis 2) each produce monotonic dose-response relationships with factual retention. The regime transition therefore cannot be attributed to adapter rank alone. The next section examines how these regimes manifest in the distributional properties of the synthetic outputs.

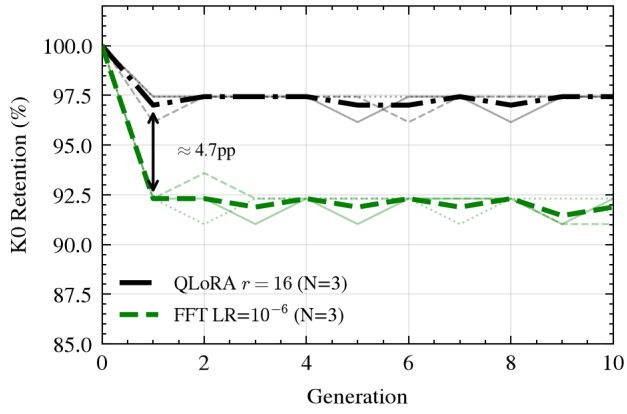


Figure 4: K0 retention over 10 recursive generations for QLoRA $r = 16$ and FFT at $LR = 10^{-6}$ on Qwen 2.5 1.5B (three seeds each). The retention gap emerges at Gen1 and remains constant through Gen10; both methods exhibit similarly stable post-adaptation trajectories.

4.4. Distributional signatures of degradation

The previous sections identified regime transitions through factual retention alone. This section asks how these regimes manifest in the distributional properties of the synthetic outputs. We apply the output-distribution diagnostics defined in §3.6 to three representative Qwen configurations: $r = 16$ (homeostatic), $r = 128$ (bounded retention), and $r = 256$ (degradative).

Homeostatic regime ($r = 16$). In the homeostatic regime, all distributional metrics remain stable across ten generations. Mean response length increases only slightly (2.47 to 2.70 words), Distinct-1 remains above 0.63, MTLD stays above 2000, stopword ratio remains near 0.16, and baseline response persistence holds steady at approximately 36%. Content efficiency declines modestly from 0.41 to 0.35, reflecting the initial one-generation adaptation cost but no progressive drift.

Bounded regime ($r = 128$): a retention/distribution dissociation. The $r = 128$ configuration reveals a dissociation between factual retention and output-distribution quality. While K0 retention remains bounded at 88.6% through Gen10, the output distribution degrades substantially: mean response length inflates from 2.47 to 7.0 words, content efficiency collapses from 0.41 to 0.13 (a threefold dilution), stopwords double from 12% to 25%, MTLD collapses from 2673 to 745, and baseline persistence falls to 5.2%. We refer to this as a *filler verbosity* phenotype: the model pads responses with repetitive function words while retaining enough factual content to maintain bounded retention.

This dissociation has a practical implication: K0 retention alone does not fully capture the onset of distributional degradation. A system monitored solely by factual accuracy would classify $r = 128$ as stable, while its generated outputs have already degraded substantially in efficiency, diversity, and response anchoring.

Degradative regime ($r = 256$). At $r = 256$, nearly all metrics diverge from their baseline values. Mean response length increases from 2.47 to 3.6 words, content efficiency declines from 0.41 to 0.22, and baseline response persistence collapses from 33% to 3.8% by Gen10. Notably, MTLD *increases* (2673 to 3804) and stopword ratio *decreases* (12% to 9%), indicating that the model generates more varied vocabulary while losing factual anchoring. We refer to this as an *elaborative drift* phenotype: the model produces longer, lexically diverse, but factually unanchored responses.

The two above-threshold configurations therefore exhibit distinct distributional failure modes despite both departing from homeostatic behavior: $r = 128$ degrades through filler accumulation, while $r = 256$ degrades through elaborative dispersion.

Three-regime taxonomy. These observations motivate a descriptive three-regime taxonomy:

1. **Homeostatic** ($r \leq 64$ on Qwen, retention above approximately 90%): factual retention and output-distribution metrics are both stable.
2. **Distributionally degraded, factually bounded** ($r = 128$ on Qwen): factual retention remains bounded while output-distribution metrics degrade substantially.
3. **Factually degradative** ($r = 256$ on Qwen): factual retention declines progressively and output distribution drifts.

This taxonomy summarizes the observed behavior of the evaluated configurations rather than defining universally distinct categories. Figure 5 illustrates the divergent trajectories across these three regimes.

Temporal relationship between output drift and retention. To examine whether output drift precedes factual loss, we computed cross-lag correlations between mean response length at generation t and retention at generation $t+1$. The raw Pearson correlation in the degradative regime is strong ($\rho = -0.965$), but robustness analyses substantially attenuate this association. After conditioning on shared temporal trend (partial correlation), the coefficient drops to -0.22 . First-difference analysis yields -0.43 , and the reverse-lag correlation is of comparable magnitude, indicating a bidirectional association rather than clear temporal precedence.

We therefore interpret output drift as a co-symptom of the same underlying high-pressure process, rather than a demonstrated causal antecedent of factual loss. Output drift remains valuable as a diagnostic signature: it is consistently present in degradative and distributionally degraded configurations, consistently absent in homeostatic ones, and can appear even when factual retention remains bounded, as illustrated by the $r = 128$ configuration. Determining the causal relationship between output-distribution changes and factual degradation remains an open question.

The next section tests whether the third component of effective training pressure, synthetic data exposure, can shift the system back toward homeostasis at the regime boundary.

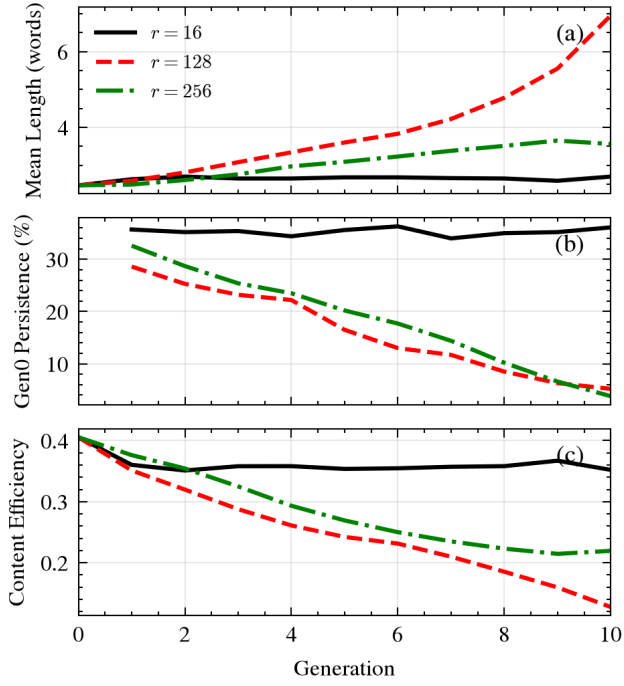


Figure 5: Distributional diagnostics over ten recursive generations for three representative Qwen configurations. (a) Mean response length, (b) baseline response persistence, (c) content efficiency. All three metrics diverge for $r = 128$ and $r = 256$ while remaining stable for $r = 16$; the $r = 128$ configuration is notable for strong distributional drift despite bounded factual retention.

4.5. Synthetic exposure: pressure sensitivity at the boundary

The preceding sections established that update capacity and perturbation magnitude each govern the regime transition. This section tests the third component of effective training pressure: whether reducing synthetic data exposure can shift the system back from a degradative to a near-homeostatic configuration. All interventions are conducted on Qwen $r = 256$, which sits at the boundary between the bounded and degradative regimes.

Intervention results. Table 5 summarizes K0 retention at Gen5 for each intervention condition relative to the unmodified C1 baseline.

Table 5

Synthetic-exposure interventions on Qwen $r = 256$. C3 is evaluated with three training seeds (15, 137, 256); C5 is evaluated with three independent random downsampling masks. Token counts refer to the total synthetic training stream per generation. Means are reported for multi-seed or multi-mask conditions.

Condition	Gen5 Ret.	Δ vs C1	Tokens	Seeds/Masks
C1 Normal	83.3% (65/78)		~57k	1
C2 Short	91.0% (71/78)	+7.7 pp	~60k	1
C3 Filtered	92.3% (mean)	+9.0 pp	~54k	$N=3$
C4 Canonical	88.5% (69/78)	+5.1 pp	~52k	1
C5 Random	92.7% (mean)	+9.4 pp	~54k	$N=3$ masks

All four interventions improve retention relative to the C1 baseline. The strongest effects are observed for C3 (length-filtered, +9.0 pp) and C5 (C3-matched random downsampling, +9.4 pp), both of which shift the system from the degradative range (83.3%) to near-homeostatic levels (92–93%).

C3 versus C5: exposure sensitivity, not quality selection. The critical comparison is between C3 and C5. C3 removes examples with long responses (those exceeding 5 words), while C5 removes a random subset of examples to approximately match C3’s token budget. Despite removing largely disjoint sets of examples (Jaccard overlap between removed sets: 0.015), both produce nearly identical improvements. Per-seed C3 Gen5 retention values are 73/78, 71/78, and 72/78 (mean 72.0/78, 92.3%); per-mask C5 values are 72/78, 73/78, and 72/78 (mean 72.3/78, 92.7%). The consistency across three independent random masks confirms that the stabilization effect does not depend on which specific examples are removed.

These results suggest that in the Qwen $r = 256$ boundary configuration, the system is sensitive to marginal changes in synthetic exposure volume rather than to the particular quality characteristics of the removed examples. Removing approximately 5% of the training stream under either evaluated selection strategy shifts the configuration toward near-homeostatic retention.

Token-budget controls. C2 (short-answer constrained) provides an important additional control. It retains all 2,000 training examples and uses slightly more total tokens than C1 (~60k vs ~57k), yet still improves retention by 7.7 percentage points. This is inconsistent with a simple token-budget explanation in which fewer training tokens mechanically reduce overfitting: the improvement persists even when total token count increases. The token difference between C1 and C3/C5 is only approximately 5% (57k vs 54k), further indicating that the effect operates near a sharp boundary rather than through gradual token reduction. Token count alone therefore cannot explain the observed recovery.

Item-level analysis. At the item level, the C1 baseline loses about eight K_0 facts between Gen0 and Gen5. C3 loses only one additional fact beyond the initial Gen1 adaptation; C5 shows a similar pattern. The interventions therefore do not merely delay degradation; they substantially reduce the number of facts displaced during the recursive process.

Figure 6 visualizes the retention and effect-size comparison across conditions.

These results demonstrate that the Qwen $r = 256$ configuration lies near a sharp regime boundary: a reduction of approximately 5% in synthetic exposure volume is sufficient to restore near-homeostatic retention in this boundary case. This sensitivity is tested only at the $r = 256$ boundary on Qwen and should not be interpreted as a universal property of recursive fine-tuning. Whether comparable boundary sensitivity exists at other ranks, on other backbones, or with different datasets remains untested.

Taken together, the intervention experiments show that the degradative regime is reversible under the evaluated conditions. Adjusting synthetic exposure restores near-homeostatic behavior in the boundary configuration, complementing the capacity and perturbation results presented in the preceding sections.

The five subsections above establish a coherent empirical picture of recursive fine-tuning under synthetic data. Increasing adapter rank produces a monotonic transition from homeostatic to degradative behavior on Qwen, while experiments on Gemma 3 show that the same qualitative regime structure generalizes across backbones despite architecture-dependent thresholds. Full fine-tuning learning-rate sweeps reproduce the same dose-response relationship, indicating that the transition is not specific to low-rank adaptation. Above-threshold configurations exhibit characteristic distributional signatures, including a dissociation between factual retention and output quality at intermediate capacity. Finally, synthetic-exposure interventions demonstrate that boundary configurations can be shifted back toward near-homeostatic behavior through modest reductions in recursive exposure. The following section interprets these findings in the context of prior work, discusses their implications for recursive training systems, and outlines the limitations of the present study.

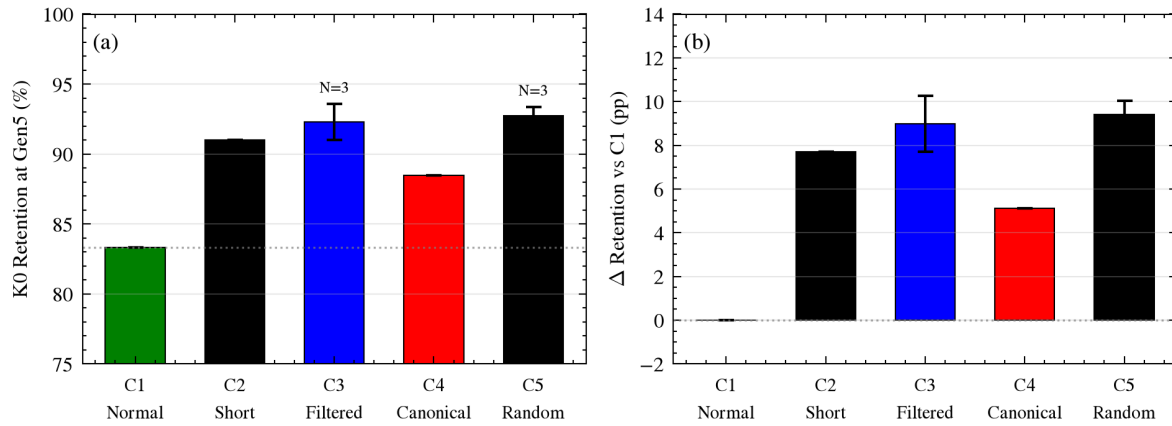


Figure 6: Intervention comparison on Qwen $r = 256$ at Gen5. (a) K0 retention by condition. (b) Retention improvement relative to C1 baseline. Error bars for C3 and C5 reflect the range across three seeds or masks.

5. Discussion

5.1. Effective training pressure as an organizing framework

5.2. Relation to recursive-training literature

5.3. Distributional degradation as a diagnostic signal

5.4. Practical implications for recursive training systems

6. Limitations

7. Conclusion

CRedit authorship contribution statement

Julio Azancort: Conceptualization, Methodology, Software, Investigation, Writing - Original Draft, Visualization.

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